

Event-Driven Programming

Definition:

Event-driven programming is a programming paradigm where the flow of the program is determined by events — such as user actions (clicks, key presses), sensor outputs, or messages from other programs/threads.

Instead of running in a strict sequence, the program waits and responds when an event occurs.

Key Concepts in Event-Driven Programming

1. **Events** – Actions or occurrences that the program reacts to (e.g., button clicked, file opened).
 2. **Event Handlers** – Methods/functions that define what should happen when an event occurs.
 3. **Event Loop** – Continuously listens for events and dispatches them to the right handlers.
 4. **Publish/Subscribe Model** – Objects can *publish* events, and others can *subscribe* to handle them.
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Where It's Used

- GUI applications (Windows Forms, WPF, Web apps – button clicks, text changes).
 - Game development (keyboard/mouse input, collisions).
 - IoT devices (sensor triggers, network messages).
 - Asynchronous programming (network requests completing, timers).
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Event-driven programming = programs that react to things happening, instead of running step by step.